

6(c)

$$(i) \text{ Specificity (True Negative)} = \frac{TN}{TN + FP}$$
$$= \frac{9560}{9560 + 140}$$

$$= 98.56\%$$

$$(ii) \text{ Sensitivity (Recall)} = \frac{TP}{FN + TP}$$

$$= \frac{90}{90 + 210} = 30\%$$

$$(iii) \text{ Overall Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= \frac{90 + 9560}{90 + 9560 + 140 + 210}$$

$$= \frac{9650}{10000}$$

$$= 96.5\%$$

- Significance of confusion matrix: A confusion matrix shows how well a classification model performs by displaying the counts of correct and incorrect predictions across all classes.

3(c) Total Patients = 1000

Actual cancer = 100

$\therefore$ Actual No cancer = $1000 - 100 = 900$

Predicted cancer = 120 of which 80 have actually

Cancer. TP = 80 and FP = 40

$\therefore$ Actual cancer not detected = $100 - 80 = 20$ (FN)

$\therefore$ TN = $900 - 40 = 860$

TP	80	20	FN
40		860	TN
FP			

(i) Precision and recall :

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{80}{80 + 40} = 66.7\%$$

$$\text{Recall} = \frac{TP}{TP + \text{FN}} = \frac{80}{80 + 20} = \frac{80}{100} = 80\%$$

(ii) Recall should be prioritized in a cancer screening system.

Because missing an actual cancer case (FN) is far more dangerous than a false alarm.

■ Draw the ROC Curve for the following;

Tuple#	Class	Prob
1	P	0.90
2	P	0.80
3	N	0.70
4	P	0.60
5	P	0.55
6	N	0.54
7	N	0.53
8	N	0.51
9	P	0.50
10	N	0.40

Solution:

Tuple	Class	Prob	TP	FP	True(+) rate ↑ TPR	Fals(+) rate ↑ FPR
1	P	0.90	1	0	$\frac{1}{5}=0.2$	$\frac{0}{5}=0$
2	P	0.80	2	0	$\frac{2}{5}=0.4$	$\frac{0}{5}=0$
3	N	0.70	2	1	0.4	0.2
4	P	0.60	3	1	0.6	0.2
5	P	0.55	4	1	0.8	0.2
6	N	0.54	4	2	0.8	0.4
7	N	0.53	4	3	0.8	0.6
8	N	0.51	4	4	0.8	0.8
9	P	0.50	5	4	1.0	0.8
10	N	0.40	5	5	1.0	1.0

Table : Information on
positive and negative
samples.

$$TPR = \frac{TP}{P}$$

$$FPR = \frac{FP}{N}$$

